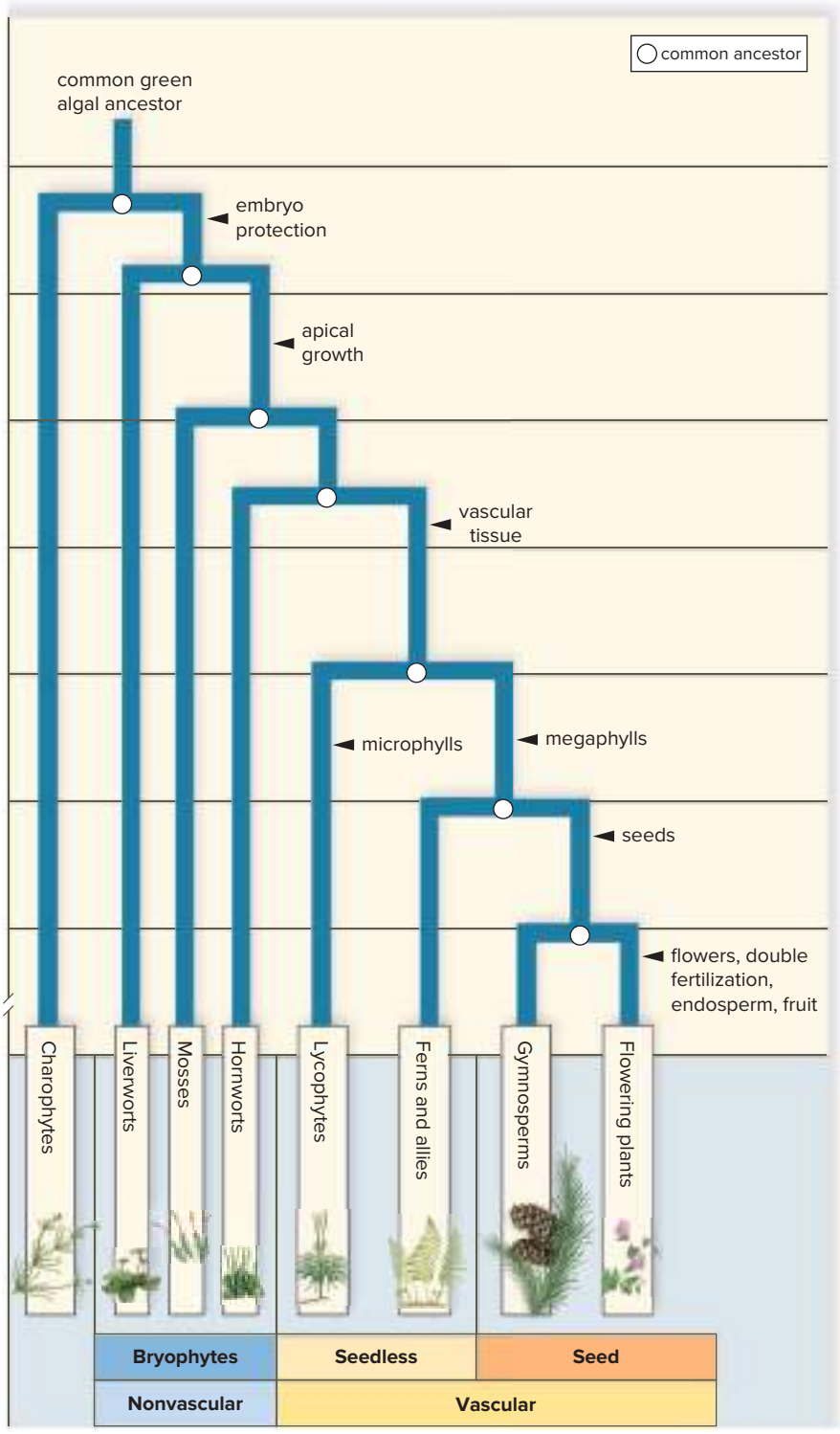


Figure 23.1 Evolutionary history of plants. The evolution of plants involves several significant innovations.

In Section 26.1, we will explore how stomata can be closed at times to limit water loss.

- Moving water within plants is a challenge that increases with plant size. Picture the height of moss compared to the height of a redwood tree. The different members of the land plants can be distinguished based on the presence or absence of **tracheids,** specialized cells with proteins that resist gravity and facilitate the upward transport of water and minerals.
- Compared to green algae, the bodies of land plants are mostly composed of three-dimensional tissues. A *tissue* is an association of many cells of the same type that can lead to specialized structures, such as organs. Tissues provide land plants with an increased ability to avoid water loss at their surfaces because bodies composed of tissues have a lower surface-area-to-volume ratio than do branched filaments, as seen in *Chara* (see Fig. 23.3).
- Plants living on land are exposed to higher intensities of UV rays than are aquatic algae. Exposure to ultraviolet light can increase the chance of mutations, but having a diploid genome can hide the effect of a single, deleterious allele. All terrestrial plants have both a haploid and a diploid generation, and the evolutionary shift toward a more dominant diploid generation allows for greater genetic variability in land plants.

Alternation of Generations

In the life cycle of humans, and other animals, the diploid stage is multicellular, whereas the haploid stage is composed of single-celled gametes (see Fig. 10.9). Plants also have a diploid and a haploid stage, but the significant difference is that both of these stages contain multicellular structures. Therefore, all land plants exhibit an **alternation of generations,** meaning that the plant has two alternating forms in the course of its life cycle. The two types of multicellular bodies are (1) the diploid, spore-producing **sporophyte** generation and (2) the haploid, gamete-producing **gametophyte** generation (Fig. 23.5).

The sporophyte produces a specialized structure called a **sporangium,** where meiosis takes place to produce haploid spores. A **spore** is a reproductive cell that develops into a new organism without the need to fuse with another reproductive cell. The spore then undergoes mitosis and becomes a new multicellular gametophyte.

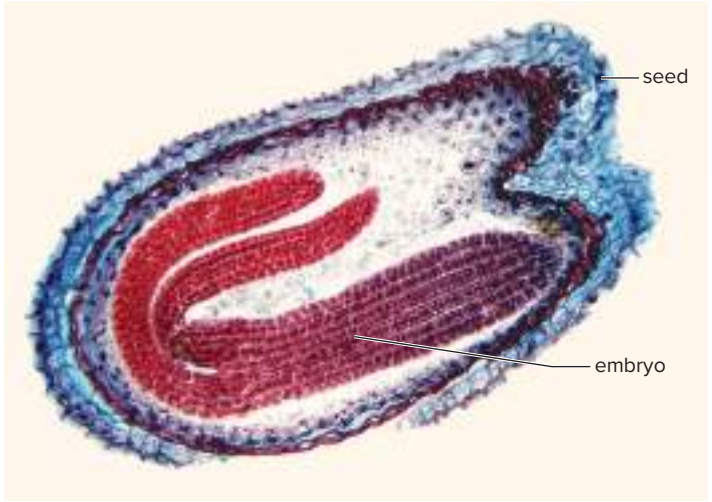
The gametophyte is a multicellular haploid structure that will develop both male and female gamete-producing regions called *gametangia.* Best seen in the moss life cycle in Figure 23.9, male gametangia are called **antheridia;** they produce sperm. The female gametangia are called **archegonia;** they produce eggs. Sperm and egg will fuse, become a *zygote,* and begin the diploid part of the life cycle.

Dominant Generation

Land plants differ as to which generation is dominant—that is, more conspicuous. In plants such as mosses, the gametophyte is



a. Embryo of the green alga, *Oedogonium* 100×



b. Embryo from the land plant, *Capsella* 60×

Figure 23.2 Embryo difference in green algae and land plants. **a.** The filamentous green alga, *Oedogonium,* with an unprotected embryo. **b.** A seed from the shepherd's purse plant, *Capsella,* showing the embryo protected and nourished. (a): Biology Pics/Science Source; (b): Garry DeLong/Science Source



Chara



Coleochaete

Figure 23.3 Charophytes. The charophytes (represented here by *Chara* and *Coleochaete*) are the green algae most closely related to the land plants. (*Chara*): Natural Visions/Alamy Stock Photo; (*Coleochaete*): Dr. Charles F. Delwiche, University of Maryland

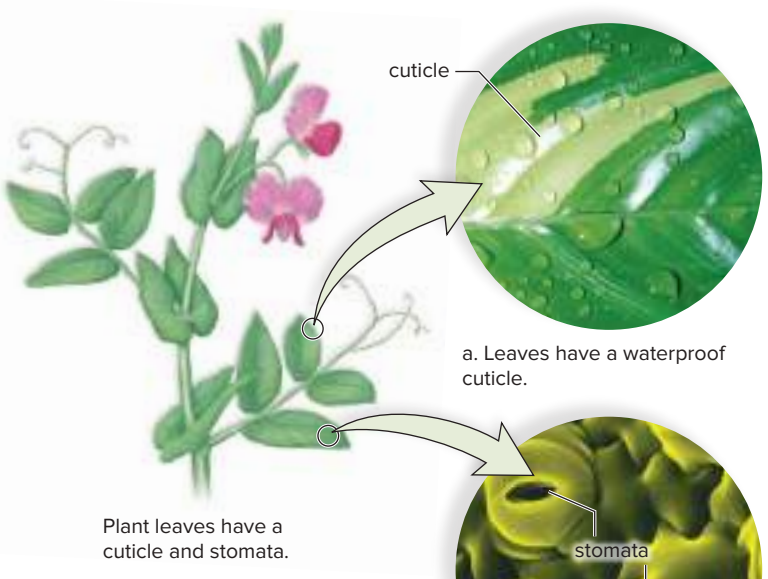


Figure 23.4 Leaf adaptation to prevent desiccation. **a.** A cuticle keeps the underlying cells and tissues from drying out. **b.** The exchange of gases is possible because the cuticle is interrupted by stomata. (a): Ingram Publishing/age fotostock; (b): Dr Jeremy Burgess/Science Source

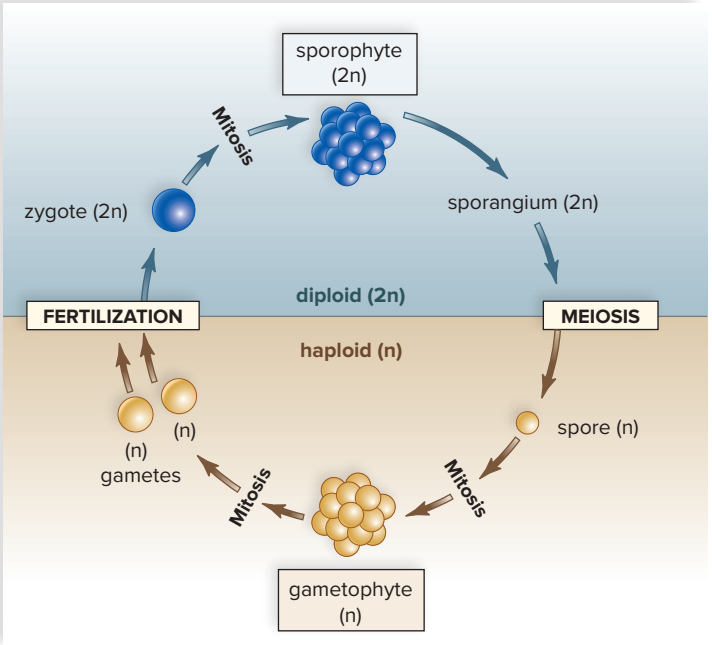


Figure 23.5 Alternation of generations in land plants. The zygote develops into a multicellular 2n generation, and meiosis produces spores in multicellular sporangia. The gametophyte generation produces gametes within multicellular gametangia.

dominant, but in plants such as ferns, pine trees, and peach trees, the sporophyte is dominant (Fig. 23.6). In the history of land plants, a vascular system evolves only in the sporophyte; therefore, the shift to sporophyte dominance is an adaptation to life on land.

Notice that as the sporophyte gains in dominance, the gametophyte becomes microscopic. The gametophyte also becomes dependent on the sporophyte.

ISTUDY